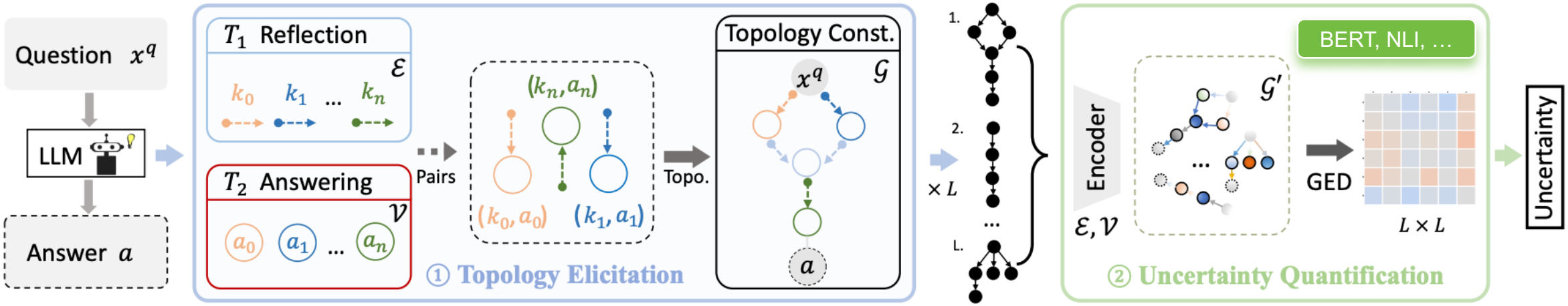
Topo-UQ

Topo-UQ[22]



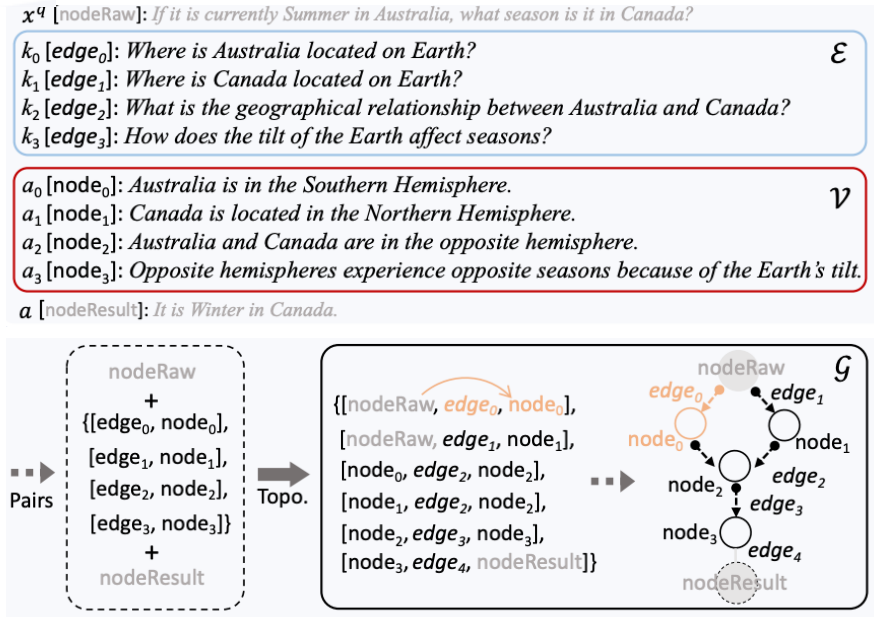
☐ Step 1: Topology Elicitation

Given an input question x^q, the model first generates sub-questions k_n, and then produces corresponding answers, forming sub-question–sub-answer pairs (k_n, a_n), where sub-questions and sub-answers are respectively treated as edges and nodes; the model further outputs dependency structures of the form $[node_i, edge_{ij}, node_j]$, organizing these steps into a reasoning topology G.

☐ Step 2: Uncertainty Quantification

given multiple reasoning topologies for the same query, the model encodes nodes and edges into semantic embeddings and computes pairwise differences using a graph edit distance variant, where uncertainty is defined as the variance of these distances, capturing both semantic and structural inconsistency.

e.g.,



[22] Understanding the Uncertainty of LLM Explanations: A Perspective Based on Reasoning Topology, COLM 2025

Topo-UQ[22]

- **Task Type:** Opening-end/Math/Code generation

- **Introduction:** Topo-UQ constructs reasoning topologies as graphs and computes uncertainty via semantic-aware graph edit distances across multiple explanations, representing reasoning through structured graph forms. It quantifies reasoning uncertainty and provides a structured perspective for analyzing consistency and variance in reasoning processes.

Type	Uncertainty Sources	Model Accessibility	Training Necessity	Consumption	Evaluation
Consensus-based	Reasoning	Black-box	Post-hoc	Medium	Ranking

[22] Understanding the Uncertainty of LLM Explanations: A Perspective Based on Reasoning Topology, COLM 2025